

Improved physics-informed neural network for reactive transport modeling of groundwater arsenic enrichment

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Abstract The enrichment and migration of geogenic high-arsenic contaminated groundwater are controlled by complex hydrological conditions and hydrogeochemical processes. Conventional reactive transport models encounter dual challenges, including poor convergence and low computational efficiency, when simulating multi-pathway reaction networks. To address these limitations, this study develops an improved physics-informed neural network (PH-PINNs) by integrating physical constraints with the PHREEQC geochemical module. The proposed model explicitly accounts for microbially mediated reaction networks involving iron-sulfur-carbon-nitrogen cycles, which drive arsenic enrichment and mobilization in groundwater. Leveraging field monitoring data from the Shanyin experimental site in the Datong Basin, Shanxi Province, a two-dimensional numerical model was constructed to simulate site-scale arsenic reactive transport. Results indicate that the PH-PINNs model significantly outperforms traditional PINNs in capturing nonlinear and non-stationary dynamics, reducing the root mean square error (RMSE) by over 50%. The model exhibits superior predictive accuracy, numerical stability, and adaptability for complex, multi-component reaction networks. This framework provides a robust tool for advancing both theoretical research and practical management of arsenic-contaminated groundwater systems.

Keywords Geogenic contaminated groundwater, Reactive network, Numerical methods, Microbially mediated, PH-PINNs

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1. Introduction

Within the complex hydrogeological and hydrogeochemical conditions, the water-rock interactions involving the parent rock and the aquifer matrix result in the formation of groundwater that is both enriched in arsenic and of diminished quality (Lone et al., 2021; Lightfoot et al., 2022), thereby threatening the health and security of human and ecological systems (Podgorski and Berg, 2020). The genesis of arsenic-rich groundwater and the migration transforma-

tion of its toxic components arise from the prolonged interaction of hydrodynamic and hydrochemical forces. This complex process involves multiple reaction pathways, including the reductive dissolution of arsenic-bearing iron oxides (Pal et al., 2002; Fendorf et al., 2010), the competitive adsorption of bicarbonate and phosphate ions (Gao et al., 2020b; Neidhardt et al., 2021), the desorption of arsenic under alkaline conditions (Gao et al., 2020a; Nguyen et al., 2022; Raju, 2022), and the release of arsenic within sulfate-reducing environments (Wang et al., 2021b; Nghiem et al., 2023). Arsenic migration is complexly connected to the

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geochemical evolution of groundwater systems (Xie et al., 2013). As aquifer conditions change, the groundwater environment gradually becomes more alkaline and reducing, promoting arsenic dissolution and release from mineral sources (Islam et al., 2004; Guo et al., 2011; Xie et al., 2013; Lin et al., 2019). Furthermore, the enrichment and migration of arsenic in groundwater are governed by a confluence of physical, chemical, and biological factors within the aquifer, including pH, arsenic speciation, adsorptive substrates, colloids, organic matter, microbial activity, and coexisting ions. These elements exert a nonlinear influence on the release, migration, and transformation of arsenic through complex networks of nonlinear adsorption and redox reactions (Dixit and Hering, 2003; Gao et al., 2020b; Stolze et al., 2022). The interplay among these factors further increases the profoundly nonlinear character of arsenic enrichment and migration processes in groundwater. Within this complex reaction network, microbially mediated processes in the iron-sulfur-carbon-nitrogen system control critical mechanisms such as adsorption-desorption, oxidation-reduction, and surface complexation (Ayotte et al., 2003; Islam et al., 2004; Wang et al., 2021a; Xiu et al., 2022).

Over the past three decades, a large number of reactive transport numerical programs and software have been developed (Prommer et al., 2003; Langevin and Guo, 2006; Liang et al., 2012), widely used to simulate the reactive solute transport processes in groundwater. Widely utilized tools include PHREEQC, PHT3D, HP1/HPx, HYTEC, eSTOMP, HYDROGEOCHEM, MIN3P, PFLOTRAN, and DuMux. In groundwater arsenic simulation, researchers initially used PHREEQC to construct one-dimensional reactive transport models, integrating organic matter processes, iron oxide reduction, and arsenic adsorption-desorption processes (Postma et al., 2007). Furthermore, some scholars utilized PHAST, MIN3P, and PFLOTRAN to develop two-dimensional reactive transport models for arsenic (Kocar et al., 2014; Huang et al., 2018; Jakobsen et al., 2018), while coupling PHREEQC with MT3DMS enabled three-dimensional reactive transport models (Prommer et al., 2003). These efforts further incorporated organic matter degradation and iron oxide interactions (Nghiem et al., 2023; Barron et al., 2024; Liang et al., 2024). Nevertheless, conventional groundwater reactive transport modeling approaches still encounter limitations in computational convergence, efficiency, and accuracy when addressing multi-component coupling, multi-pathway reaction networks, and large-scale regional simulations (Wei and Cao, 2022; Atteia et al., 2024). Moreover, model calibration demands a large amount of data and time (Chen et al., 2020).

The complex processes of arsenic enrichment and migration in groundwater, characterized by multi-component coupling and multi-pathway reaction networks—such as Fe (III) reduction, sulfate reduction, and adsorption-desorption

—are typically represented by stiff nonlinear equation systems (Steeffel et al., 2005). Conventional Newton-Raphson iterative methods are highly sensitive to initial conditions and abrupt changes in reaction rates, often resulting in computational non-convergence or numerical oscillations (Singh et al., 2014). Traditional numerical approaches, such as finite element and finite difference methods, rely on mesh discretization, where computational complexity scales with problem size and mesh resolution. Limitations in mesh resolution can compromise the accurate representation of localized reaction zones, such as redox interfaces or adsorption bands, leading to issues like numerical dispersion that undermine computational precision (Tang et al., 2016). Moreover, the computational cost of generating high-resolution meshes increases rapidly (Mohamed et al., 2024). Additionally, chemical reaction kinetic parameters are of significant uncertainty (Zhang L et al., 2024). Model calibration requires extensive field data, and parameter estimation involves complex nonlinear optimization processes, further increasing computational complexity (Thiros et al., 2021).

In recent years, physics-informed neural networks (PINNs) have emerged as a promising approach, effectively integrating physical constraint equations into deep learning frameworks, thus offering novel perspectives for modeling complex systems in environmental science (Raissi et al., 2019, 2020; Yang et al., 2021). Unlike traditional numerical methods, PINNs optimize neural network training through physical law constraints without relying on grid discretization, making them particularly well-suited to high-dimensional, complex problems (Zobeiry and Humfeld, 2021; Ji et al., 2021; Chiu et al., 2022; Guo et al., 2023). Studies have demonstrated PINNs' robust performance in simulating groundwater flow and predicting contaminant dispersion (Ali et al., 2024; Ji et al., 2024; Secci et al., 2024; Wang et al., 2024; Zhan et al., 2024). However, PINNs exhibit limitations when tackling complex reaction systems and multi-field coupling problems, with considerable scope for enhancing computational efficiency (Chiu et al., 2022; Wu et al., 2023; Ryu et al., 2024; Zhang S et al., 2024). To address challenges in groundwater modeling, researchers have developed various PINN variants, including nonlocal physics-informed neural networks (nPINNs) for nonlocal problems (Pang et al., 2020), parallel physics-informed neural networks (PPINNs) for efficiently handling time-dependent partial differential equations (Meng et al., 2020), and conservative physics-informed neural networks (CPINNs) tailored for solving systems of equations (Jagtap et al., 2020).

In the Datong Basin, groundwater arsenic levels grossly exceed permissible limits, posing a severe carcinogenic risk to residents (He et al., 2021; Li et al., 2021; Chi et al., 2024; Li et al., 2024; Su et al., 2016). The concentration of arsenic in groundwater shows significant temporal and spatial variability, resulting from coupled hydrological, biological,

and geochemical processes. Nevertheless, the mechanisms driving these concentration fluctuations require further quantitative study (Gorny et al., 2015; Xie et al., 2016; Gao et al., 2020a, 2020b; Kumar et al., 2020; Xiao et al., 2021). This study addresses the reactive transport of groundwater within complex reaction networks by integrating physical constraints with the PHREEQC geochemical reaction module, to develop an improved physics-informed neural networks (PH-PINNs) model. Through optimized network architecture, enhanced enforcement of physical laws, and optimized training strategies, this model achieves high precision and efficiency in simulating the enrichment and reactive transport processes of arsenic in groundwater. Specifically, PHREEQC enables the simulation of chemical reactions such as arsenic dissolution-precipitation, adsorption-desorption, and oxidation-reduction (Parkhurst and Appelo, 1999), while PINNs provide efficient solutions for reactive solute transport equations for arsenic (Raissi et al., 2019; Chiu et al., 2022). Through this coupled computational approach, the model accurately simulates and predicts groundwater flow and the migration and enrichment processes of arsenic.

2. PH-PINNs simulation methodology

2.1 Physics-informed neural networks

The PINNs model is specifically designed to efficiently and accurately solve complex physical equations governing multi-component chemical transport and reactions in groundwater (Figure 1) (Raissi et al., 2019). Unlike conventional neural networks, which predominantly rely on data-driven approaches, PINNs incorporate physical laws as prior knowledge to guide network training. The key innovation lies in its loss function, which accounts not only for data mismatches but also for errors in the physical equations, yielding major improvements in both computational efficiency and precision (Hu et al., 2024). This feature is particularly useful for groundwater reactive transport problems, where data may be sparse yet the basic physical mechanisms are well-defined. Mathematically, if partial differential equations (PDEs) are expressed as:

$$F(u, \nabla u, \nabla^2 u, \dots, x, t) = 0, \quad (1)$$

where u represents the physical quantity to be solved (such as hydraulic head or concentration), with x and t denoting spatial and temporal coordinates, respectively, and ∇ signifying the gradient operator. Within the PH-PINNs framework, a multilayer neural network approximates u , denoted as $u_{NN}(x, t; \theta)$, where θ represents the network parameters. This approximate function is then inserted into the governing physical equations, with the resulting residual becoming a component of the loss function. Furthermore, the physical

solution must satisfy initial conditions (IC) and boundary conditions (BC), which correspondingly generate losses associated with IC and BC. Consequently, the loss function for PH-PINNs may be expressed as:

$$L(\theta) = \omega_1 L_{\text{data}}(\theta) + \omega_2 L_{\text{physics}}(\theta) + \omega_3 L_{\text{BC}}(\theta) + \omega_4 L_{\text{IC}}(\theta), \quad (2)$$

where $L_{\text{data}}(\theta)$ denotes the error associated with observational data, $L_{\text{physics}}(\theta)$ represents the error derived from physical equations, while $L_{\text{BC}}(\theta)$ and $L_{\text{IC}}(\theta)$ represent the constraint errors for boundary and initial conditions, respectively. The parameters ω_1 , ω_2 , ω_3 , and ω_4 are the respective dynamic weights. These loss components are combined with specific weights to form the total loss value (Cuomo et al., 2022), which is subsequently minimized through gradient descent techniques (Tian et al., 2023). Furthermore, we used a gradient magnitude-based weighting approach, dynamically adjusting the relative weights by computing the magnitude of gradient variations across different constraint terms, expressed as:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \left(\frac{\|\nabla L_j\|}{\sum_j \|\nabla L_j\|} \right)}{\sum_k \left(\frac{\|\nabla L_k\|}{\sum_j \|\nabla L_j\|} \right)}, \quad (3)$$

where $\|\nabla L_i\|$ denotes the gradient magnitude of the loss term L_i at the current training iteration t , with the normalization term ensuring that the weights sum to unity. This approach balances the contributions of data-driven and physics-constrained losses, maintaining equitable influence among diverse loss terms throughout the training process, thus enabling efficient training even in scenarios with sparse data (Raissi et al., 2019).

Scholars have integrated the physical constraints of partial differential equations (PDEs) into the architecture of deep neural networks, achieving an end-to-end numerical solution workflow (Wang and Zhong, 2024). This model directly embeds physical constraints related to multi-component reactive transport systems, such as mass conservation equations and chemical kinetic equations, into the neural network training process. Using automatic differentiation (AD) techniques, the model precisely computes the temporal derivatives and spatial gradients of solute concentrations, thereby ensuring the following of the physical constraints governing chemical transport and reactions (Ji et al., 2021).

2.2 The coupled simulation methodology of PH-PINNs

This study introduces a simulation method, called PH-PINNs, which integrates a physics-informed neural network with the geochemical reaction model PHREEQC. Using a

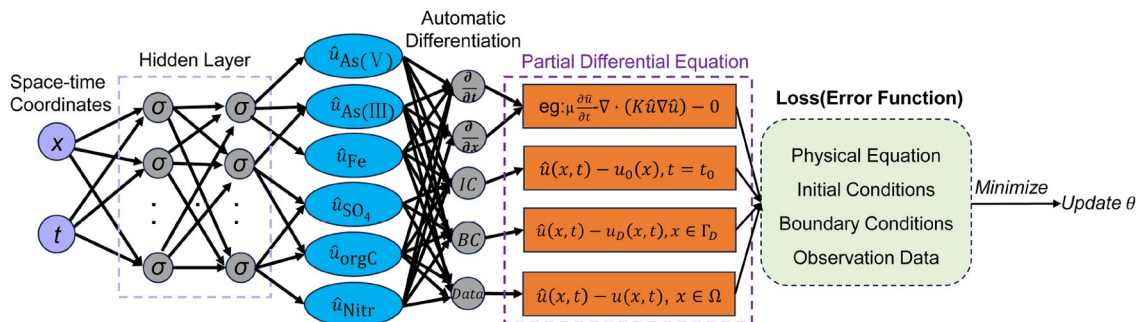


Figure 1 Schematic representation of the PINNs network architecture and solution of partial differential equations.

sequential non-iterative approach (SNIA), it achieves efficient dynamic coupling of hydrodynamics, dispersion, and chemical reactions (Figure 2). By combining the computational processes of physical fields and chemical reaction networks, this method overcomes the convergence and computational efficiency challenges in traditional approaches when solving multi-pathway reaction network systems.

Specifically, PH-PINNs use spatial and temporal coordinates (x, t) as inputs, with the neural network outputting concentrations of groundwater chemical constituents, including arsenic (As(V), As(III)), iron (Fe), sulfate (SO_4), organic carbon (orgC), and nitrate (Nitr). Compared with conventional PINNs, the PH-PINNs model has been enhanced in the following aspects.

(1) Optimization of physical constraints: By integrating the PHREEQC geochemical module, the nonlinear dynamic simulations of sulfate reduction, iron oxide dissolution, and arsenic adsorption-desorption processes were significantly enhanced.

(2) Dynamic weighting of loss functions: Addressing disparities among different components and reaction rates, Phreeqpy computations are activated by predefined concentration thresholds. The recalculated hydrochemical equilibrium states are iteratively updated as inputs for PINNs, while adaptively adjusting weight coefficients of constraint terms in the loss function (as detailed in Section 2.1). This methodology greatly improves the model's adaptability to complex geochemical systems.

(3) Enhancement of computational efficiency: Using parallel computing techniques, computational time was reduced by over 30% while preserving numerical accuracy.

The core innovations of the PH-PINNs methodology lie in: (1) Dynamic collaborative computing of physics and chemistry: PH-PINNs achieve simultaneous resolution of physical fields and chemical reaction processes through a bidirectional coupling mechanism between PINNs and PHREEQC. By embedding governing partial differential equations (PDEs) of physical processes, PINNs enable mesh-free simulation of global distributions for physical variables

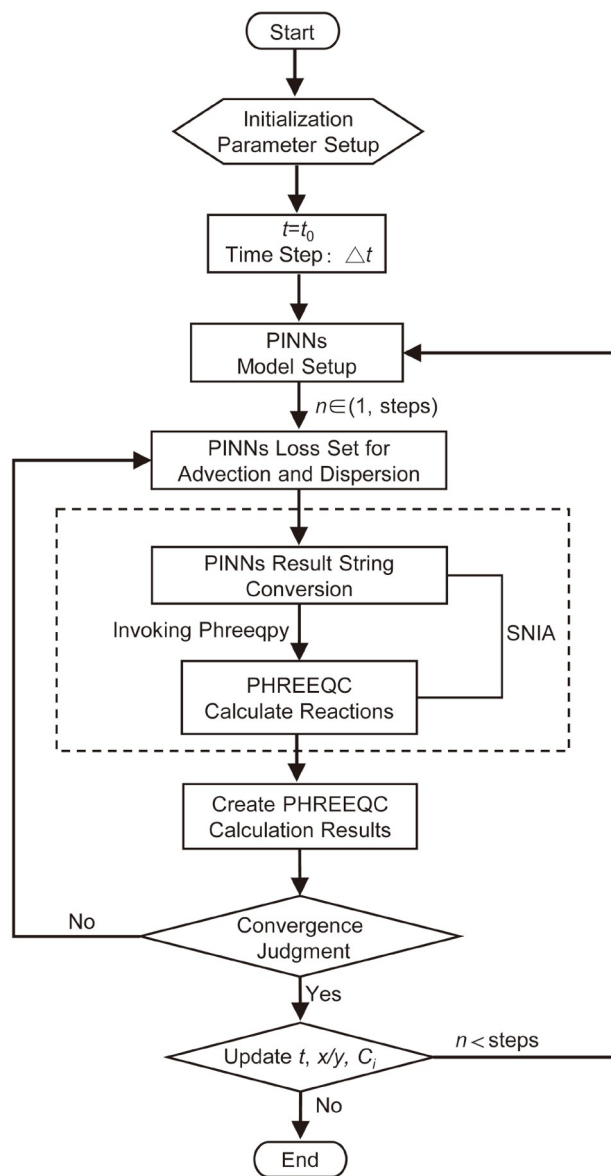


Figure 2 Workflow for numerical simulation of reactive solute transport using PH-PINNs.

(e.g., flow velocity, dispersion coefficients). PHREEQC specializes in high-fidelity computation of complex chemical

reactions, including dissolution, precipitation, and equilibrium speciation. At each temporal iteration, physical variables calculated by PINNs are dynamically transmitted to PHREEQC, which subsequently quantifies chemical constituent concentration variations under input conditions. These updated chemical states are reciprocally fed back into PINNs for physical variable refinement. This bidirectional coupling mechanism ensures physicochemical consistency and preserves numerical precision throughout simulations.

(2) Efficiency of SNIA: Distinct from conventional iterative coupling paradigms, PH-PINNs employ SNIA to independently resolve physical fields and chemical reactions, circumventing computational overhead from excessive iterations. Specifically, physical variables computed by PINNs across temporal increments directly serve as PHREEQC inputs for dynamic chemical process simulation. This non-iterative framework not only enhances computational efficiency but also mitigates numerical instability between physical and chemical domains.

(3) Implementation of adaptive convergence mechanisms: To ensure numerical stability, PH-PINNs integrate dynamic convergence evaluation at each timestep. Through real-time assessment of solution errors and convergence metrics, the system adaptively adjusts temporal increments and coupling weights for physicochemical parameters (refer to eq. (3) in Section 2.2). This self-regulating protocol guarantees both numerical stability and physical plausibility of solutions at every simulation stage.

2.3 Conceptual model

A microbially mediated iron-sulfur-carbon-nitrogen reaction network conceptual model (Figure 3) was developed to elucidate arsenic mobilization mechanisms in groundwater through accelerated iron (oxyhydr)oxide reduction and sulfate reduction. This framework couples sulfate reduction with organic carbon oxidation processes, wherein generated sulfides chemically reduce iron minerals to form mobile thioarsenic species with diminished adsorption capacity, thereby enhancing arsenic mobility. Prior studies demonstrate that sulfate reduction substantially promotes arsenic release and transport even in low-sulfate environments, resulting in aquifer arsenic exceedances (Nghiem et al., 2023). Our numerical modeling framework systematically quantifies these coupled biogeochemical dynamics through high-resolution spatiotemporal simulations.

Furthermore, arsenic enrichment involves multifaceted reaction mechanisms (principal processes summarized in Table 1), where microbial reduction of iron (oxyhydr)oxides and subsequent release of arsenic adsorbed on their surfaces are widely recognized as pivotal controls on arsenic mobilization (Islam et al., 2004; Nghiem et al., 2023). Microbial proliferation in reducing environments has been identified as the primary driver of organic matter degradation, iron oxide dissolution, and arsenic liberation. Notably, sulfate reduction accelerates arsenic mobilization even in iron oxide-rich aquifers. Current understanding posits that H_2S generated

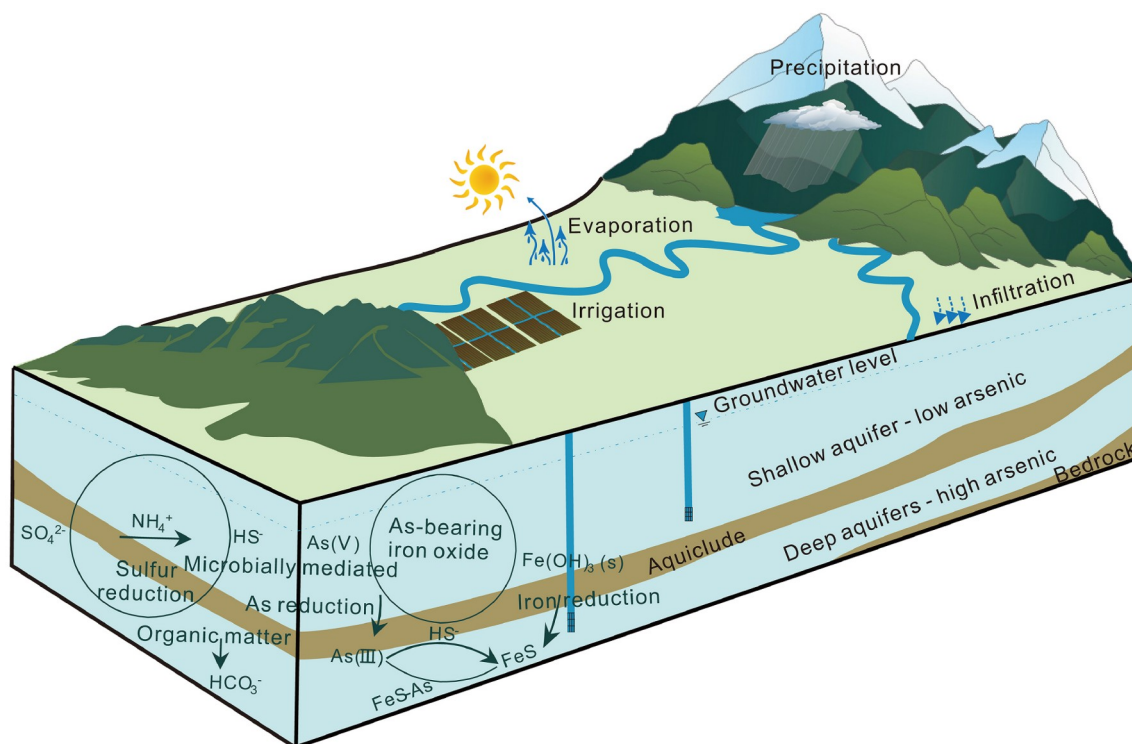


Figure 3 Conceptual model of the biogeochemical reaction network for the arsenic reaction cycle.

during sulfate reduction reduces arsenate to more mobile arsenite. In deep reducing environments, arsenic transport and enrichment are predominantly governed by sulfate reduction and weakly adsorbed thioarsenic species, with traditional adsorption-desorption controls on iron minerals contributing merely ~5% of total arsenic mobility (Nghiem et al., 2023). These processes occur concomitantly with arsenate reduction, arsenic oxidation, organic carbon mineralization, and surface complexation reactions.

The reaction rates presented in Table 1 are delineated in Table 2. The kinetic rates of Fe(III) reduction, arsenic release, sulfate reduction, and the growth of sulfate-reducing bacteria are elucidated using the Monod equation, incorporating an inhibition term to account for the suppressive effect of arsenic on microbial activity. Furthermore, the growth rate of sulfate-reducing bacteria is contingent not only upon their inherent maximum growth potential but also upon the availability of organic carbon, sulfate, and nitrogen, as well as the influence of other inhibitory factors.

The maximum growth rates, denoted as $\mu_{\max}^{\text{IRB}}$, $\mu_{\max}^{\text{SRB}}$, $\mu_{\max}^{\text{ARB}}$, and $\mu_{\max}^{\text{COB}}$, correspond to iron-reducing bacteria (IRB), sulfate-reducing bacteria (SRB), arsenate-reducing bacteria (ARB), and organic carbon-oxidizing bacteria (COB), respectively. The term X_{bio} represents the microbial biomass inhibition factor, which modulates the suppressive effect of bacterial concentration on growth rates. The biomass concentrations of IRB, SRB, ARB, and COB are expressed as Y_{IRB} , Y_{SRB} , Y_{ARB} , and Y_{COB} , respectively. The concentrations and half-saturation constants for arsenate, organic carbon, Fe(OOH), nitrogen, sulfate, dissolved oxygen, and arsenite are represented as $C_{\text{As(V)}}$, C_{OrgC} , $C_{\text{Fe(III)}}$, C_{Nitr} , C_{SO_4} , C_{O_2} , $C_{\text{As(III)}}$ and $K_{\text{As(V)}}$, K_{OrgC} , $K_{\text{Fe(III)}}$, K_{Nitr} , K_{SO_4} , K_{O_2} , $K_{\text{As(III)}}$, respectively. The inhibition constant for oxygen competition and the oxygen concentration are denoted as K_{O_2} and C_{O_2} , respectively. To reduce the number of unknown model parameters, it is assumed that the parameters K_{OrgC} and K_{O_2} remain

Table 1 Processes and equations governing arsenic reactions^{a)}

Number	Mechanisms and processes	Biogeochemical reaction
1	The microbial-mediated reductive dissolution of iron oxides	$\text{CH}_2\text{O} + 3.72[\text{Sediment-Fe(OH)}_3 - 0.01\text{As(V)}] + 0.02\text{NH}_4^+ + 6.64\text{H}^+ \rightarrow 0.9\text{HCO}_3^- + 3.72\text{Fe}^{2+} + 0.0372\text{As(V)} + 0.02\text{C}_5\text{H}_7\text{O}_2\text{N} + 9.42\text{H}_2\text{O}$
2	Sulfate reduction coupled with organic matter decomposition	$\text{CH}_2\text{O} + 0.02\text{NH}_4^+ + 0.45\text{SO}_4^{2-} \rightarrow 0.9\text{HCO}_3^- + 0.02\text{C}_5\text{H}_7\text{O}_2\text{N} + 0.45\text{H}_2\text{S} + 0.02\text{H}^+ + 0.06\text{H}_2\text{O}$
3	Arsenate reduction	$2\text{HASO}_4^{2-} + \text{CH}_2\text{O} + \text{H}^+ \rightarrow 2\text{H}_2\text{AsO}_3^- + \text{HCO}_3^-$
4	Arsenic speciation	$\text{As(III)} + 0.5\text{O}_2 \xrightarrow{k_3} \text{As(V)}$
5	Aerobic respiration	$\text{CH}_2\text{O} + \text{O}_2 \rightarrow \text{HCO}_3^- + \text{H}^+$

a) $\text{C}_5\text{H}_7\text{O}_2\text{N}$ represents the microbial biomass of iron-reducing bacteria (IRB) and sulfate-reducing bacteria (SRB), derived through kinetic modeling; $[\text{Sediment-Fe(OH)}_3 - 0.01\text{As(V)}]$ denotes arsenic adsorbed onto the surface of iron oxides within the sediment; CH_2O signifies dissolved organic carbon; As(V) and HASO_4^{2-} indicate arsenate; H_2AsO_3^- refers to arsenite.

Table 2 Rates of chemical reactions

Number	Biogeochemical reaction
1	$R_1 = \mu_{\max}^{\text{IRB}} \cdot X_{\text{bio}} \cdot \frac{K_{\text{As(V)}}}{K_{\text{As(V)}} + C_{\text{As(V)}}} \cdot \frac{C_{\text{OrgC}}}{K_{\text{OrgC}} + C_{\text{OrgC}}} \cdot \frac{C_{\text{Fe(III)}}}{K_{\text{Fe(III)}} + C_{\text{Fe(III)}}} \cdot \frac{C_{\text{Nitr}}}{K_{\text{Nitr}} + C_{\text{Nitr}}} \cdot Y_{\text{IRB}}$
2	$R_2 = \mu_{\max}^{\text{SRB}} \cdot X_{\text{bio}} \cdot \frac{C_{\text{OrgC}}}{K_{\text{OrgC}} + C_{\text{OrgC}}} \cdot \frac{C_{\text{SO}_4}}{K_{\text{SO}_4} + C_{\text{SO}_4}} \cdot \frac{C_{\text{Nitr}}}{K_{\text{Nitr}} + C_{\text{Nitr}}} \cdot Y_{\text{SRB}}$
3	$R_3 = \mu_{\max}^{\text{ARB}} \cdot X_{\text{bio}} \cdot \frac{C_{\text{OrgC}}}{K_{\text{OrgC}} + C_{\text{OrgC}}} \cdot \frac{C_{\text{As(V)}}}{K_{\text{As(V)}} + C_{\text{As(V)}}} \cdot \frac{K_{\text{O}_2}}{K_{\text{O}_2} + C_{\text{O}_2}} \cdot Y_{\text{ARB}}$
4	$R_4 = k_3 C_{\text{As(III)}} C_{\text{O}_2}$
5	$R_5 = \mu_{\max}^{\text{COB}} \cdot X_{\text{bio}} \cdot \frac{C_{\text{OrgC}}}{K_{\text{OrgC}} + C_{\text{OrgC}}} \cdot \frac{C_{\text{O}_2}}{K_{\text{O}_2} + C_{\text{O}_2}} \cdot Y_{\text{COB}}$

consistent within the equations (Huang et al., 2018). Parameters (such as $K_{As(V)}$, K_{SO_4} , $K_{Fe(III)}$, and K_{Nitr}) are derived from the literature and validated through laboratory results.

To simplify the model, all reactive iron oxides in aquifer sediments are presumed to be hydrous ferric oxides (HFO) with equilibrium surface complexation constants. Considering the potential competitive inhibition of arsenic surface complexation by carbonate species in groundwater, this study also accounts for carbonate adsorption. Reaction parameters are sourced from the literature. Table 3 presents the surface complexation reactions and equilibrium constants employed in the model.

3. Simulation of arsenic migration and transformation process

3.1 Study area

3.1.1 Hydrogeological context

The Datong Basin, situated at the northern extremity of the Shanxi Rift System, constitutes a Cenozoic fault-bounded basin (see Figure 4). Owing to its distinctive tectonic characteristics and geological milieu, this region exhibits exceptionally propitious conditions for arsenic enrichment (Li et al., 2014; Li et al., 2024; Yan et al., 2024). The basin's sediments predominantly comprise medium to coarse sand and clay, with layers enriched in arsenic-bearing minerals and soluble arsenic, fostering a dynamic chemical environment conducive to arsenic migration and transformation. The simulation area was designated at the Shanyin Xiaogeda test site, where a two-dimensional saturated unsteady groundwater flow model was developed. Data were derived from continuous manual monitoring conducted be-

Table 3 Surface complexation reactions and parameters^{a)}

Reaction	LogK
$\text{Surf}_{\text{OH}} + \text{H}^+ = \text{Surf}_{\text{OH}_2^+}$	7.29 ^{a)}
$\text{Surf}_{\text{OH}} = \text{Surf}_{\text{O}^-} + \text{H}^+$	8.93 ^{a)}
$\text{Surf}_{\text{OH}} + \text{H}^+ + \text{AsO}_4^{3-} = \text{Surf}_{\text{AsO}_4^{2-}} + \text{H}_2\text{O}$	18.10 ^{b)}
$\text{Surf}_{\text{OH}} + 2\text{H}^+ + \text{AsO}_4^{3-} = \text{Surf}_{\text{HASO}_4^-} + \text{H}_2\text{O}$	24.43 ^{b)}
$\text{Surf}_{\text{OH}} + 3\text{H}^+ + \text{AsO}_4^{3-} = \text{Surf}_{\text{H}_2\text{AsO}_4} + \text{H}_2\text{O}$	29.88 ^{b)}
$\text{Surf}_{\text{OH}} + \text{H}^+ + \text{AsO}_3^{3-} = \text{Surf}_{\text{AsO}_3^{2-}} + \text{H}_2\text{O}$	30.01 ^{c)}
$\text{Surf}_{\text{OH}} + 2\text{H}^+ + \text{AsO}_3^{3-} = \text{Surf}_{\text{HASO}_3^-} + \text{H}_2\text{O}$	31.87 ^{b)}
$\text{Surf}_{\text{OH}} + 3\text{H}^+ + \text{AsO}_3^{3-} = \text{Surf}_{\text{H}_2\text{AsO}_3} + \text{H}_2\text{O}$	32.08 ^{b)}
$\text{Surf}_{\text{OH}} + \text{H}^+ + \text{CO}_3^{2-} = \text{Surf}_{\text{CO}_3^-} + \text{H}_2\text{O}$	12.78 ^{d)}
$\text{Surf}_{\text{OH}} + 2\text{H}^+ + \text{CO}_3^{2-} = \text{Surf}_{\text{HCO}_3^-} + \text{H}_2\text{O}$	20.37 ^{d)}

a) Dzombak and Morel, 1991; b) Dixit and Hering, 2003; c) Stollenwerk et al., 2007; d) Appelo et al., 2002.

tween December 2010 and November 2011, utilizing 20 horizontally distributed nested wells (see Figure 5a, 5b), with the simulation domain encompassing dimensions of 75 m×30 m×22 m.

3.1.2 Distribution of arsenic in groundwater

The sediments within the study area exhibit elevated arsenic concentrations, ranging from 10.29 to 46.92 mg kg⁻¹, with an average of 22.71 mg kg⁻¹. Analysis of sediment samples reveals that those with anomalously high arsenic content

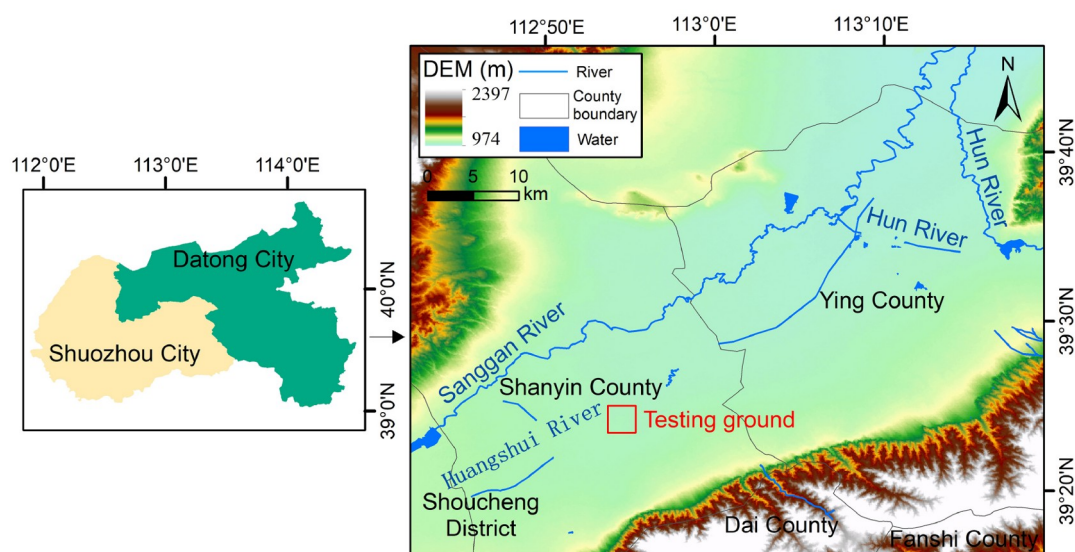


Figure 4 Location map of the study area and test site.

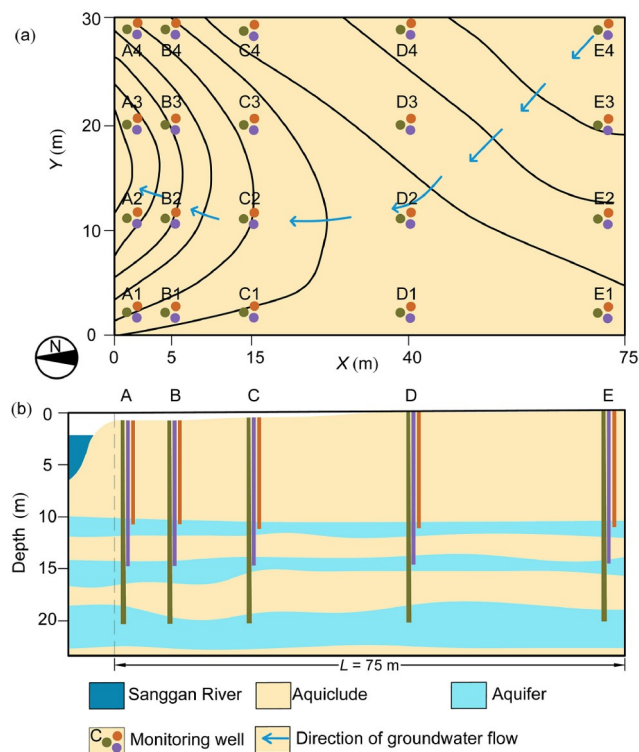


Figure 5 (a) Plan view of the study site and well placement; (b) cross-sectional profile of the site.

consistently display relatively elevated levels of iron and total organic carbon (TOC), underscoring the pivotal role of iron-bearing minerals and clay sediments in arsenic sequestration at the monitoring site.

Groundwater arsenic concentrations at the test site vary from 3 to 527 $\mu\text{g L}^{-1}$. The highest average arsenic concentration in groundwater, reaching 494 $\mu\text{g L}^{-1}$, was recorded at a depth of 19.8 m in borehole D1, surpassing 526 $\mu\text{g L}^{-1}$, while the lowest average, 9 $\mu\text{g L}^{-1}$, was observed at 14.7 m in borehole D2. Notably, 86% of the samples exceeded the World Health Organization's recommended drinking water limit of 10 $\mu\text{g L}^{-1}$. High-arsenic zones are predominantly confined to depths between 19 and 22 m below the surface. Furthermore, groundwater arsenic concentrations exhibit seasonal fluctuations, displaying an overall upward trend from March to November 2011, with particularly pronounced increases in the deeper aquifer layers.

3.2 Establishment of the reactive transport model

This study seeks to develop a two-dimensional reactive transport model using PH-PINNs to simulate the migration and transformation of arsenic in groundwater, with a particular emphasis on elucidating the spatial distribution patterns of arsenic across aquifers at varying depths and the coupled mechanisms governing its interaction with hydrodynamic conditions and chemical reactions. The upper and lower

boundaries of the simulated confined aquifer are designated as impermeable, while all four lateral boundaries are assigned prescribed head conditions. The aquifer medium consists predominantly of medium to coarse sand, with clay serving as the impermeable medium. The lithological distribution is relatively uniform, and the two-dimensional hydraulic conductivity and porosity of the porous medium are informed by empirical values. The vertical hydraulic conductivity is set at 0.1 times the horizontal value and serves as the initial parameter for model calibration.

The model domain is conceptualized into six geological units (see Figure 5b), comprising three heterogeneous, anisotropic aquifers at distinct depths, interspersed with impermeable layers. These aquifers are located at depths of 11.0, 14.7, and 19.8 m below the surface, corresponding to shallow, middle, and deep layers, respectively. The domain is discretized using a uniform grid of 150 rows (x -direction), 60 columns (y -direction), and 44 layers (z -direction), yielding a total of 396,000 cells with a resolution of 0.5 m. Table 4 consolidates the mathematical model parameters discussed in Table 2, while Table 5 delineates the hydraulic parameters and physical characteristics of the aquifers corresponding to the numerical model.

The measured hydrochemical constituents, including arsenic concentrations, are incorporated as chemical boundary conditions along the perimeter of the model domain. Hydraulic head values at observation wells on the corresponding boundaries are linearly interpolated, with groundwater level data from December 2010 serving as the initial head. Initial concentrations of arsenic species and other hydrochemical substances in the groundwater are derived from field monitoring data (see Table 6). Furthermore, the groundwater temperature in the study area remains consistently around $(10.0 \pm 2.0)^\circ\text{C}$ throughout the year, exerting minimal influence on the rates of primary chemical reactions (Jin and Bethke, 2007). Consequently, a constant temperature of 10°C is adopted for the simulation calculations.

4. Results and discussion

4.1 Simulation and analysis of groundwater flow and arsenic distribution

Groundwater in the study area flows from the high-head region in the southern part of the model domain toward the low-head region in the north, with PH-PINNs simulations demonstrating robust spatiotemporal responses in flow velocity and groundwater level variations. The spatial distribution of flow velocity exhibits pronounced heterogeneity, with the medium to coarse sand aquifer displaying high permeability and a maximum flow rate of approximately 0.35 m d^{-1} , while the clay layer exhibits minimal flow, at merely 0.001 m d^{-1} . Additionally, groundwater levels re-

Table 4 Principal parameters in the mathematical model

Parameter	Value	Parameter	Value
$K_{As(V)}$ (mol L ⁻¹)	1.9×10^{-3} ^{e)}	μ_{max}^{IRB} (d ⁻¹)	1.656 ^{e)}
$K_{As(III)}$ (mol L ⁻¹)	3.1×10^6 ^{e)}	μ_{max}^{SRB} (d ⁻¹)	2.976 ^{d)}
K_{OrgC} (mol L ⁻¹)	2.89×10^{-4} ^{j)}	μ_{max}^{ARB} (d ⁻¹)	1.0 ^{e)}
$K_{Fe(III)}$ (mol L ⁻¹)	5.2×10 ^{h)}	μ_{max}^{ARB} (d ⁻¹)	10 ^{g)}
K_{Nitr} (mol L ⁻¹)	1.00×10^7 ⁱ⁾	ν_{decay} (d ⁻¹)	2.64 ^{g)}
K_{SO_4} (mol L ⁻¹)	5.00×10^5 ⁱ⁾	X_{max} (d ⁻¹)	10 ^{b)}
K_{O_2} (mol L ⁻¹)	2.41×10^5 ^{j)}	X_{min} (d ⁻¹)	0.001 ^{b)}
k_3 (mol (L s) ⁻¹)	3×10^{-3} ^{h)}	X_{bio} (d ⁻¹)	1.0 ^{b)}

b) Estimated value; c) Wichlacz and Unz, 1985; d) Dev et al., 2016; e) Soda et al., 2006; f) Kinzelbach et al., 1991; g) Hu, 2008; h) Huang et al., 2018; i) Prommer et al., 2007; j) Zarnetske et al., 2012.

Table 5 Hydraulic parameters and physical characterization of the aquifer

Parameter	Shallow	Middle	Deep
Effective porosity	0.35	0.33	0.25
Horizontal permeability coefficient (m d ⁻¹)	20	15	10
Vertical permeability coefficient (m d ⁻¹)	5	5	2
Specific storage coefficient (m d ⁻¹)	0.0001	0.0001	0.0001
Specific yield	0.22	0.22	0.22
Horizontal dispersion coefficient (m ² s ⁻¹)	1.0×10^{-9}	1.0×10^{-9}	1.0×10^{-9}
Vertical dispersion coefficient (m ² s ⁻¹)	1.0×10^{10}	1.0×10^{10}	1.0×10^{10}

spond significantly to seasonal fluctuations (see Figure 6).

Regarding arsenic concentration distribution, PH-PINNs simulations (Figure 7) reveal that low-velocity stagnation zones are critical for arsenic enrichment, particularly in the central and deeper parts of the model domain, where concentrations progressively increase over time, ultimately forming distinct high-concentration zones. Localized peak concentrations reach 0.5 mg L⁻¹, far exceeding the national standard of 10 µg L⁻¹. Model accuracy, assessed using root mean square error (RMSE) as defined in Equation 4, indicates that PH-PINNs predictions align closely with the spatial distribution trends of monitoring data, yielding an RMSE of 3.536 µg L⁻¹, compared with 17.123 µg L⁻¹ for the PINNs model.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{i,p} - C_{i,a})^2}, \quad (4)$$

where $C_{i,p}$ is the simulated predicted concentration, $C_{i,a}$ is the actual observed concentration, and n is the total observed quantity.

Vertically, arsenic concentrations exhibit a gradual upward

trend, with notable fluctuations in the deep aquifer at depths of 20 to 22 m. This pattern may stem from weaker hydrodynamic conditions in deeper layers, which restrict discharge, coupled with adsorption-desorption processes in chemical reactions. Furthermore, seasonal variations between dry and wet periods significantly influence arsenic distribution: during the dry season (May), arsenic concentrations range from 2.8 to 46.3 µg L⁻¹, whereas in the wet season (October), the range widens to 3.5 to 181.5 µg L⁻¹, underscoring the controls of seasonal groundwater flow in arsenic migration and accumulation.

4.2 Influence of hydrological and geochemical interactions on arsenic reactive transport

The simulation results (Figures 7–10) show that PH-PINNs achieves high accuracy in modeling sulfate reduction, iron reductive dissolution, and arsenic adsorption-desorption processes, matching well with observed monitoring data. Sulfate is reduced to sulfide through the activity of sulfate-reducing bacteria (SRB), concurrently releasing significant

Table 6 Chemical composition of initial and boundary conditions

Parameter	Shallow layer	Middle layer	Deep layer	Northern boundary (Sanggan River)	Southern boundary
T ($^{\circ}\text{C}$)	11.8	8.2	9.5	8.8	10.1
pH	8.44	7.02	7.86	7.63	7.84
Eh (mV)	37	−255.4	−115.1	−56.6	−16.8
TDS (mg L^{-1})	13178	1108	5636	—	—
NH_4^+ (mg L^{-1})	4.24	0.05	1.06	0.64	0.59
Fe^{2+} (mg L^{-1})	2.68	0.01	0.29	0.03	0.03
Ca^{2+} (mg L^{-1})	785.5	27.38	241.3	386.10	251.58
Mg^{2+} (mg L^{-1})	699.4	48.57	261.4	277.1	333.9
Na^+ (mg L^{-1})	2750	171.3	1212	999.3	1107.0
HCO_3^- (mg L^{-1})	1074	152.4	509.5	336.7	430.4
SO_4^{2-} (mg L^{-1})	2945	22.7	1208	1966	1900
Cl^- (mg L^{-1})	4454	171.1	2152	1800	1728
As ($\mu\text{g L}^{-1}$)	341.1	1.42	104.5	22.65	24.15
Organic matter (mol L^{-1})	3.14	5.11	17.73	6.17	2.58

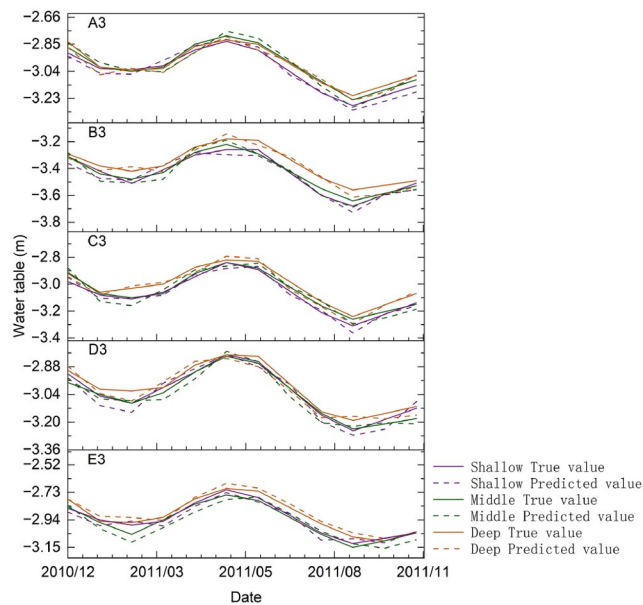


Figure 6 Time-series plot of observed and simulated groundwater levels (Wells A3, B3, C3, D3, E3 are positioned at distances of 0, 5, 20, 40, and 75 m from the riverbank, respectively).

quantities of arsenic (Figure 8). Notably, during the initial phase of declining sulfate concentrations, arsenic levels rise rapidly, indicating that sulfate reduction serves as a primary driver of arsenic release.

Iron oxides, as key carriers of arsenic within the aquifer, play a critical role in its immobilization and release through redox reactions. In discharge zones, where reducing condi-

tions intensify, the dissolution rate of iron oxides significantly increases, leading to large amounts of Fe(II) and arsenic entering the aqueous phase (Figure 9). Simulations reveal that Fe(II) concentrations in the middle aquifer significantly exceed those in the deeper aquifer, reflecting depth-dependent variations in iron oxide dissolution. This suggests that iron reduction has less impact on arsenic immobilization than sulfate reduction.

The study further highlights that the redox potential of groundwater is a key factor controlling arsenic speciation and mobility (Figure 10). Under oxidizing conditions, arsenic predominantly exists as pentavalent As(V), less mobile; conversely, under reducing conditions, trivalent As(III) prevails, displaying significantly enhanced mobility. The transformation of arsenic species is complicated linked to the spatial distribution of hydrodynamic conditions. The PH-PINNs simulations of redox potential and arsenic concentration distributions exhibit a high degree of concordance, affirming their capacity to accurately replicate the complex dynamic behavior of groundwater chemistry.

4.3 Performance of the improved PH-PINNs model

Analysis of curves and root mean square error reveals that the PH-PINNs model substantially outperforms traditional PINNs in complex chemical reaction systems (Figure 11). PH-PINNs effectively captures sharp nonlinear fluctuations in concentration peaks for total arsenic (As_{tot}), iron (Fe^{2+}), and bicarbonate (HCO_3^-), while accurately replicating the

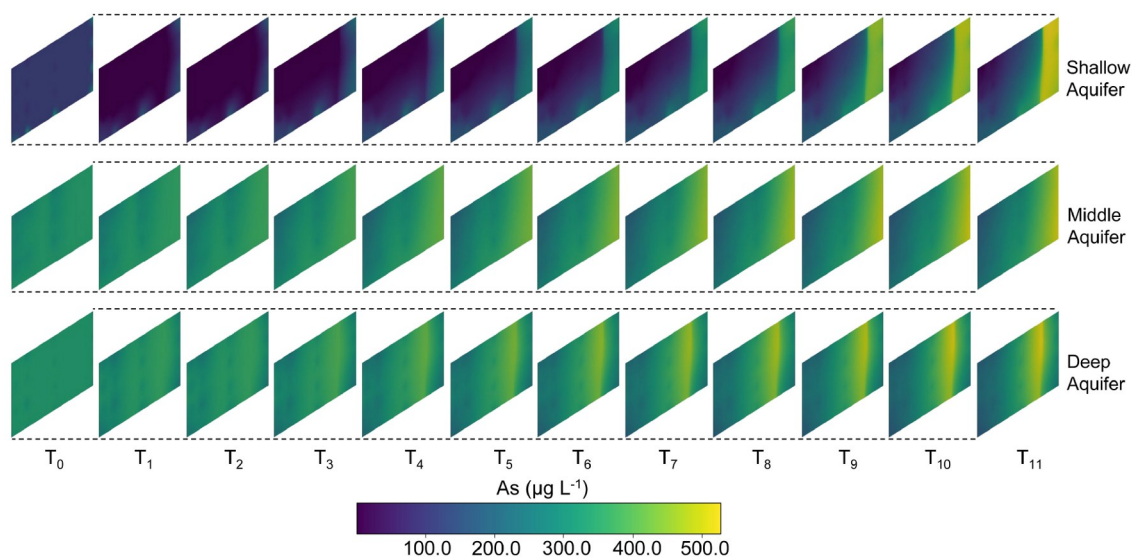


Figure 7 Simulated temporal evolution of arsenic concentrations in different aquifers (T_0 – T_{11} denote time steps with a monthly interval, $\Delta T=1$ month, where T_0 corresponds to December 2010 and T_{11} to November 2011; cross-sections represent horizontal profiles of shallow, middle, and deep aquifers at depths of 11.0, 14.7, and 19.8 m below the surface, respectively, as applicable to Figures 8–10).

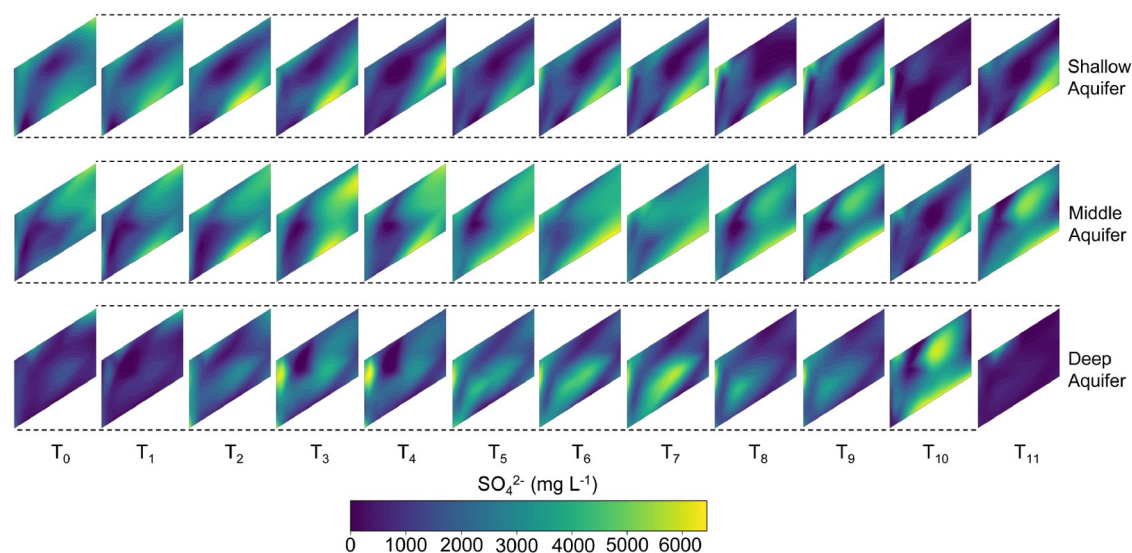


Figure 8 Simulated temporal evolution of SO_4^{2-} concentrations in horizontal profiles of different aquifers.

dynamic interplay between sulfate reduction and arsenic release. Across predictions for nine ionic components, PH-PINNs consistently achieves lower root mean square errors than traditional PINNs, with an average reduction exceeding 50%, demonstrating superior stability and adaptability in multi-component chemical reaction systems. For instance, under the complex reducing conditions of the deep aquifer, traditional PINNs show large errors in sulfate concentration and arsenic speciation predictions, whereas PH-PINNs precisely captures these trends. The inverse variation of SO_4^{2-} and the gradual rise of HCO_3^- are effectively modeled by PH-PINNs, while conventional PINNs fall short in reducing

environments and reactive transport scenarios.

Overall, by integrating physical constraints and coupled chemical reaction modules, PH-PINNs significantly enhances both accuracy and applicability, offering a robust tool for simulating multi-component, highly nonlinear systems. In contrast, while traditional PINNs boast higher computational efficiency, their precision and applicability in complex reaction systems are limited.

5. Conclusion

This study combines physical constraints with the

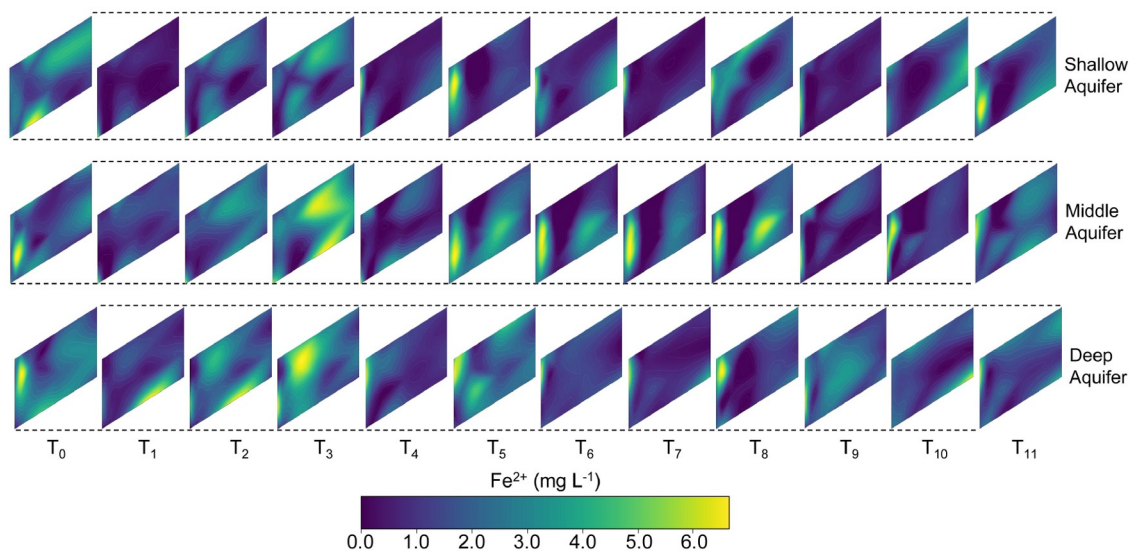


Figure 9 Simulated temporal evolution of Fe^{2+} concentrations in horizontal profiles of different aquifers.

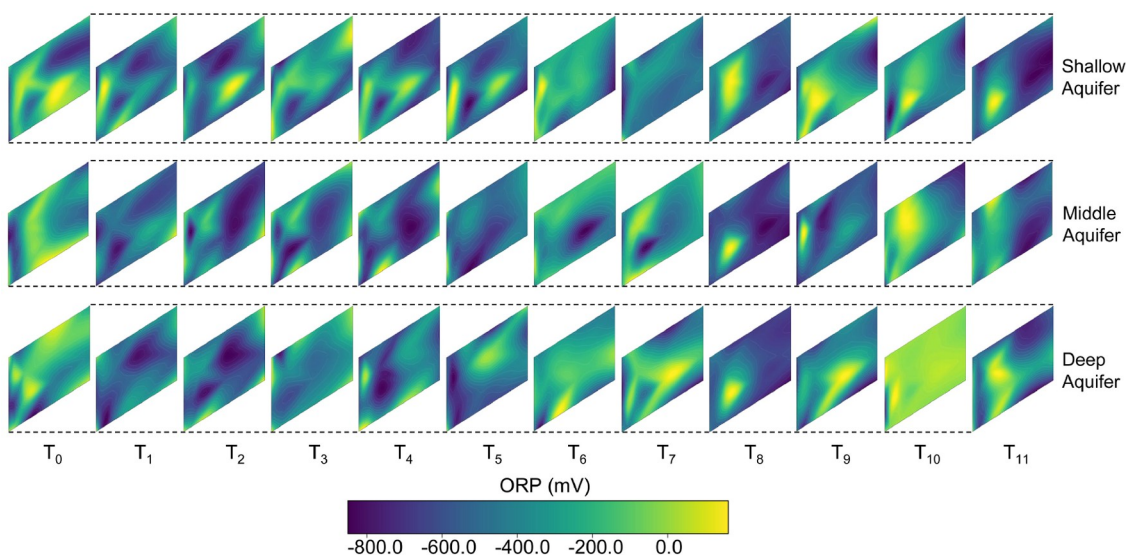


Figure 10 Simulated temporal evolution of oxidation-reduction potential (ORP) in horizontal profiles of different aquifers.

PHREEQC geochemical reaction module to develop an improved Physics-Informed Neural Network model (PHPINNs). The model establishes a site-scale 2D numerical framework for reactive transport, effectively simulating arsenic enrichment and migration processes in groundwater. It quantifies the spatiotemporal distribution patterns of arsenic migration and transformation within a microbial-mediated iron-sulfur-carbon-nitrogen reaction network governing groundwater arsenic enrichment. The principal findings are as follows.

(1) The strong spatial heterogeneity of flow velocity in the study area significantly influences arsenic migration pathways and enrichment zones. Low-velocity flow zones emerge as critical areas for arsenic accumulation, with par-

ticularly notable enrichment in deeper layers, where maximum concentrations exceed 0.5 mg L^{-1} , surpassing national groundwater quality standards.

(2) The enrichment and transformation of arsenic in groundwater are controlled by the combined effects of hydrodynamic conditions and redox environments. Sulfate reduction, iron oxide dissolution, and arsenic adsorption-desorption processes are complicated, coupled, dictating arsenic release, accumulation, and speciation. Changes in oxidation-reduction potential (ORP) directly modulate arsenic's chemical forms, with trivalent arsenic (As(III)) becoming the major species under low ORP conditions, exhibiting significantly enhanced mobility.

(3) By integrating physical constraints with complex

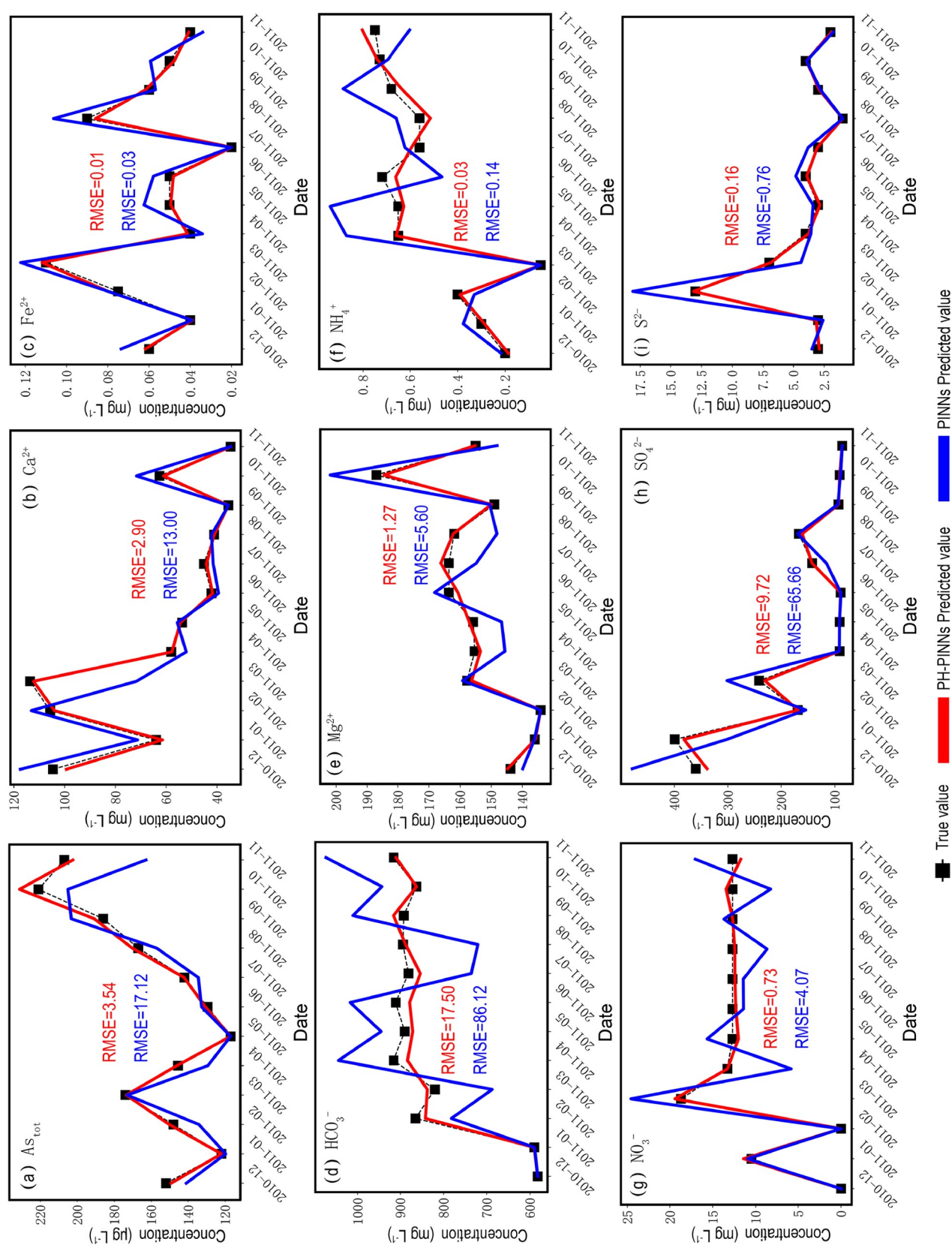


Figure 11 Simulation results for multi-component PINNs and PH-PINNs.

geochemical reaction networks, the PH-PINNs model substantially enhances the accuracy and computational efficiency of simulating groundwater arsenic enrichment and migration. The root mean square error of PH-PINNs predictions is over 50% lower than that of conventional PINNs, demonstrating superior stability and adaptability. This framework holds promise for further application in long-term, regional-scale numerical simulations of groundwater arsenic enrichment under three-dimensional conditions.

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Conflict of interest The authors declare that they have no conflict of interest.

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